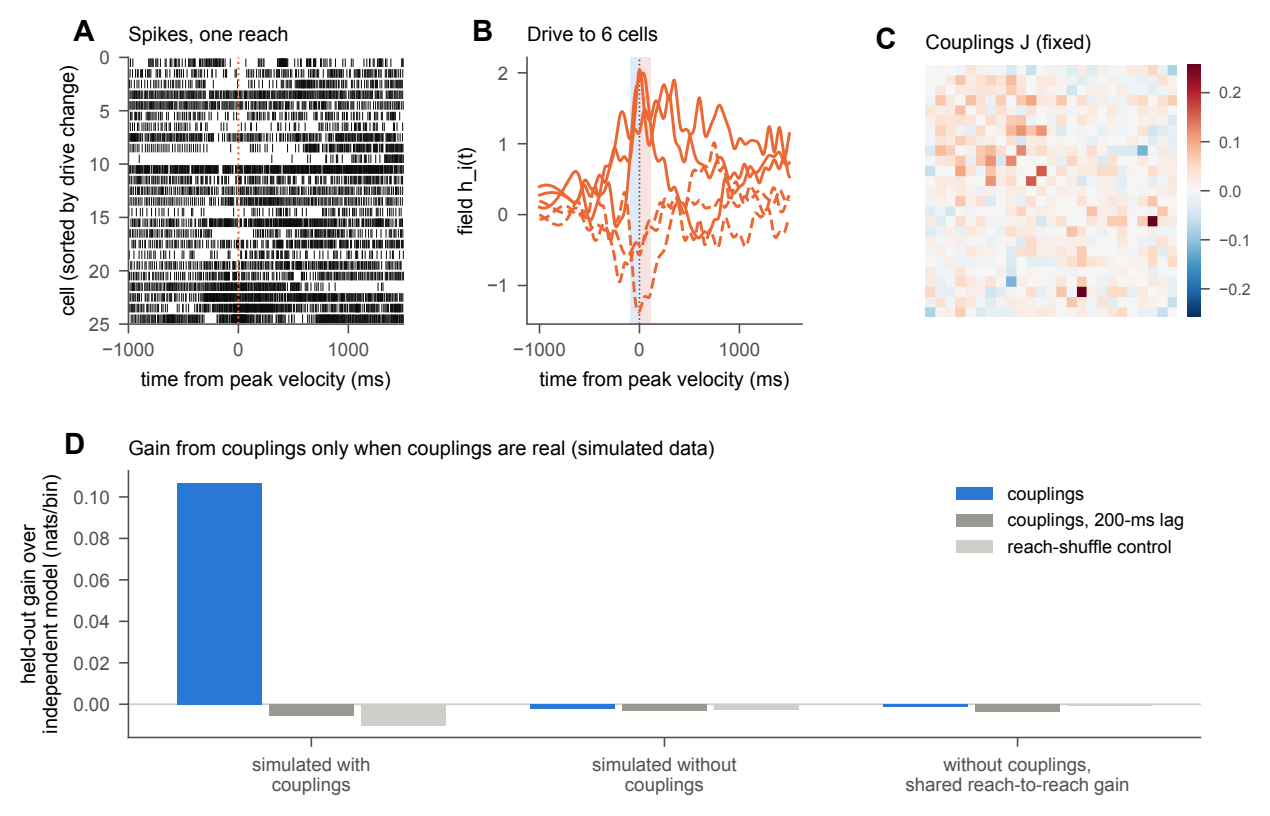

A
Spikes, one reach
cell (sorted by drive change)
0
5
10
15
20
25
−1000
0
1000
time from peak velocity (ms)
B
Drive to 6 cells
field h_i(t)
2
1
0
−1
−1000
0
1000
time from peak velocity (ms)
C
Couplings J (fixed)
0.2
0.1
0.0
−0.1
−0.2
D
Gain from couplings only when couplings are real (simulated data)
held-out gain over independent model (nats/bin)
0.10
0.08
0.06
0.04
0.02
0.00
couplings
couplings, 200-ms lag
reach-shuffle control
simulated with couplings
simulated without couplings
without couplings, shared reach-to-reach gain